

Business Problem Statement

This project addresses the need for a retail analytics solution that can turn a large transactional dataset into clear, actionable business insights. The dataset contains over one million retail transactions and includes information about customers, cities, store types, payment methods, seasons, products, revenue, and transaction behavior. The core business challenge is not just to analyze the data, but to make it understandable for decision-making in a practical and visual way.

Business Problems to Solve

1. Understand retail performance across different regions and store formats.
 - Which cities and store types are driving the most revenue?
 - Which combinations perform best or worst?
2. Identify customer and transaction patterns.
 - How do transactions differ by customer category, season, and store type?
 - Which customer segments are generating the most value?
3. Evaluate revenue and profitability trends.
 - How does revenue change over time?
 - Which periods show growth, decline, or seasonality?
4. Assess payment behavior and shopping patterns.
 - Which payment methods are most commonly used?
 - Are there differences in customer behavior by store type or city?
5. Create a simple decision-support dashboard.
 - Stakeholders should be able to explore the data interactively without writing code.
 - The dashboard should highlight key KPIs and trends clearly.

Business Goals

The project aims to:

- clean and prepare the raw retail data for analysis,
- perform exploratory data analysis (EDA) to uncover meaningful patterns,
- design a set of business-friendly KPIs similar to grocery and retail performance metrics,
- build an interactive dashboard for monitoring business performance,
- provide a reusable analytics workflow that can be shared with others through GitHub and Streamlit.

Expected Outcome

By the end of the project, the business should be able to use the dashboard to quickly understand retail performance, compare cities and store types, identify high- and low-performing segments, and make informed operational and strategic decisions.